

RAFFLES INSTITUTION
2017 Year 6 Preliminary Examination

Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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INDEX
NUMBER

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BIOLOGY

Paper 4 Practical

9744/04

25th August 2017

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name on all the work you hand in.

Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen on both sides of the paper.

You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **questions 1 and 2** in the **spaces** provided on the **Question Paper**.

Answer **question 3** in the **foolscap paper** provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part of the question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
Total	

This document consists of **16** printed pages.



Raffles Institution
Internal Examination

Answer **all** the questions.

- 1** Many plant cells contain a water soluble molecule called ascorbic acid, which has many functions to help the plant survive.

You are provided with an extract from plant cells, **P**, which contains ascorbic acid.

Visking tubing, **V**, is selectively permeable, similar to a cell surface membrane, so that some biological molecules will diffuse through the wall of the tubing.

You are required to investigate the diffusion of ascorbic acid from **P** into the water surrounding the Visking tubing over a period of 15 minutes.

Fig. 1.1 shows the apparatus before the water was added.

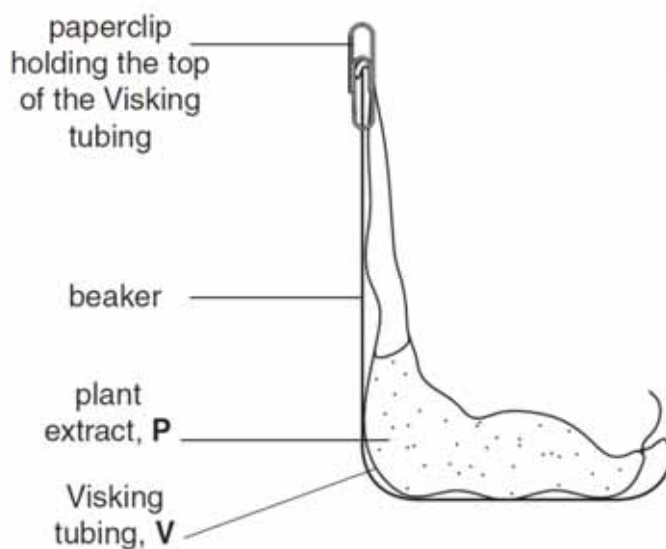


Fig. 1.1

- (a) (i)** Water is added to the beaker in Fig. 1.1.

Describe the expected trend in the concentration of ascorbic acid in the water over a period of 15 minutes.

.....
 [1]

You are provided with:

labelled	contents	hazard	volume / cm ³
A	sample of water removed after 15 minutes	irritant	15
P	plant extract containing ascorbic acid	irritant	15
W	distilled water	none	100
I	iodine in potassium iodide solution	irritant	20
S	starch	none	20

labelled	details
V	15 cm length of Visking tubing in a beaker containing water

You must now read up to the end of step 23 before proceeding.

To compare the concentration of ascorbic acid in the samples, you are required to find the **volume** of iodine solution, **I**, added to each sample until the end-point is reached. The drops of **I** will be added one at a time using a small syringe.

To practice releasing drops from a small syringe:

1. Fill the syringe with 2 cm³ of **I**.
2. Hold the syringe over an **empty** test-tube and push the plunger **gently** to release one drop at a time, as shown in Fig. 1.2.

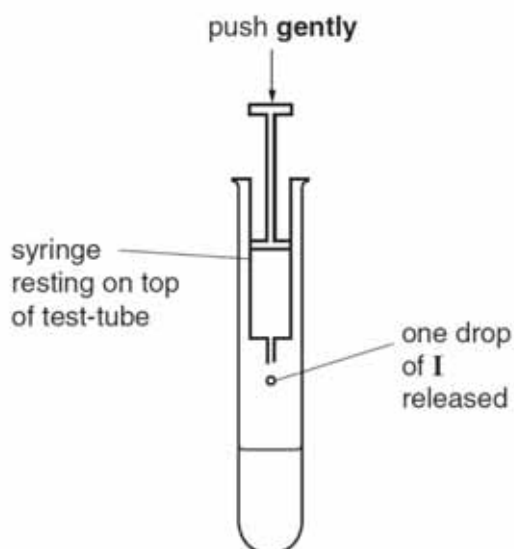


Fig. 1.2

To compare the concentration of ascorbic acid in the samples you will need to add drops of **I** until a blue colour appears. When this blue colour lasts for more than 10 seconds, this is the end-point and the **volume of I** that has been added should be recorded.

The apparatus in Fig. 1.1 was set-up and water was added and left for 15 minutes.

Sample **A** was removed from the water in the beaker.

You are required to find the volume of **I** needed to reach the end-point for sample **A**.

Proceed as follows:

3. Put 1 cm³ of **S** into a test-tube.
4. Put 3 cm³ of the sample (e.g. **A**) into the same test-tube.
5. Shake the test-tube gently to mix the contents.
6. Fill the syringe, labelled **I**, with 2 cm³ of **I**.
7. Wipe off any drops of **I** from the outside of the syringe with a paper towel.
8. Add **one** drop of **I** to the mixture in the test-tube as shown in Fig. 1.2.
9. Mix gently and if there is no colour change add another drop.
10. Continue adding drops, one at a time, until the blue colour appears.

Wait 10 seconds to see if the end-point has been reached. If the blue colour disappears then add another drop.

11. Repeat step 10 until the mixture stays blue for at least 10 seconds

(ii) Record the **volume of I** needed to reach the end-point.

volume[1]

You are required to:

- set up Visking tubing containing **P** as in Fig. 1.3
- decide the level of water to put into the beaker
- remove samples of the water surrounding the Visking tubing at **5 minute intervals** for 15 minutes
- compare the ascorbic acid concentrations in the samples.

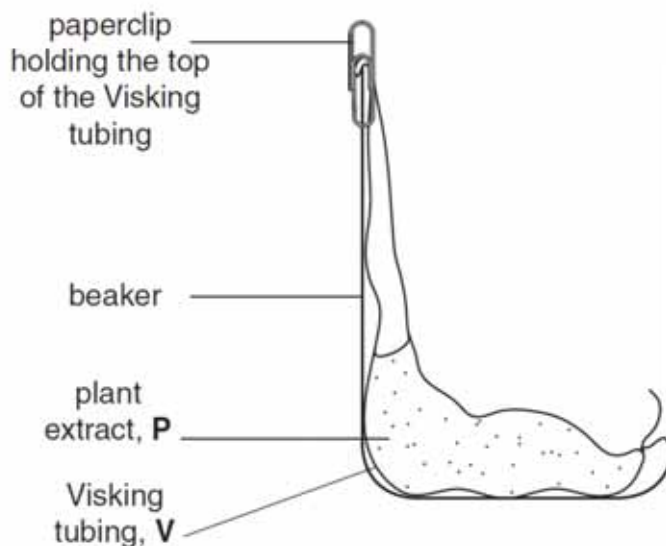


Fig. 1.3

Samples of water surrounding the Visking tubing will be removed for testing, so you need to take this into account when you decide the level of water to put into the beaker.

(iii) Draw **on** Fig. 1.3 the level of the water

- before you remove any samples (label 'before'),
- after the total volume of water needed for all the tests has been removed (label 'after').

[2]

(iv) In order to compare the ascorbic acid concentrations,

state **one** variable which you will need to standardise when finding the volume of **I** added to each sample and describe how you will standardise this variable.

variable

description.....

.....[1]

Proceed as follows:

12. Put **S**, as in **step 3**, into the four test-tubes you will require in order to test the samples of water.
13. Tie a knot in the Visking tubing as close as possible to one end so that it seals the end.
14. To open the other end, wet the Visking tubing and rub the tubing gently between your fingers.
15. Put 6 cm³ of **P** into the open end of the Visking tubing.
16. Rinse the outside of the Visking tubing by dipping it into a beaker of water.
17. Then put the Visking tubing into an empty beaker as shown in Fig. 1.3.
18. Make sure the open end of the Visking tubing is held in place by a paperclip.

You will start timing as soon as you add **W** (**steps 19 and 20**).

*You should read **steps 19 to 23** before proceeding.*

19. Put **W** into the beaker to the level you decided in **(iii)**.
 20. Immediately start timing and remove the first sample of water (as in **step 4**) and put into a prepared test-tube (as in **step 12**).
 21. Test the sample as in **steps 5 to 11**.
 22. After 5 minutes, gently mix the water surrounding the Visking tubing and then remove the next sample, put it into a different (prepared) test-tube and repeat **steps 5 to 11**.
 23. Repeat **step 22** for two more samples.
- (v)** Prepare the space below and record your results.

- (vi) Describe how the results support your expected trend as stated in (a)(i).

.....

 [1]

This investigation provides results to compare the concentration of ascorbic acid in the samples.

- (vii) The actual concentration of ascorbic acid cannot be quantified in (v).

If you had been provided with 1.0% ascorbic acid solution, suggest how you would modify this investigation to find the **percentage concentration** of ascorbic acid in the water after 15 minutes.

.....

 [3]

- (viii) A systematic error occurs when apparatus with scales are used, since the scales may be slightly different.

For example, when measuring the same line, two rulers may give different lengths.

However, as long as the same ruler is used for all the measurements, the trend is **not** affected because the error is consistent.

State **one** piece of apparatus used in this investigation that may have a systematic error. Suggest whether this affected your results and give a reason for your answer.

apparatus
 reason.....
 [1]

- (b)** Iodine solution (iodine in potassium iodide solution) turns blue-black when starch is present in plant tissues.

However, as ascorbic acid is also found in plant tissues, some scientists investigated the effect of testing for starch with iodine solution when there was ascorbic acid present.

The concentration of ascorbic acid was $0.0001 \text{ mol dm}^{-3}$ and the concentration of starch solution was standardised.

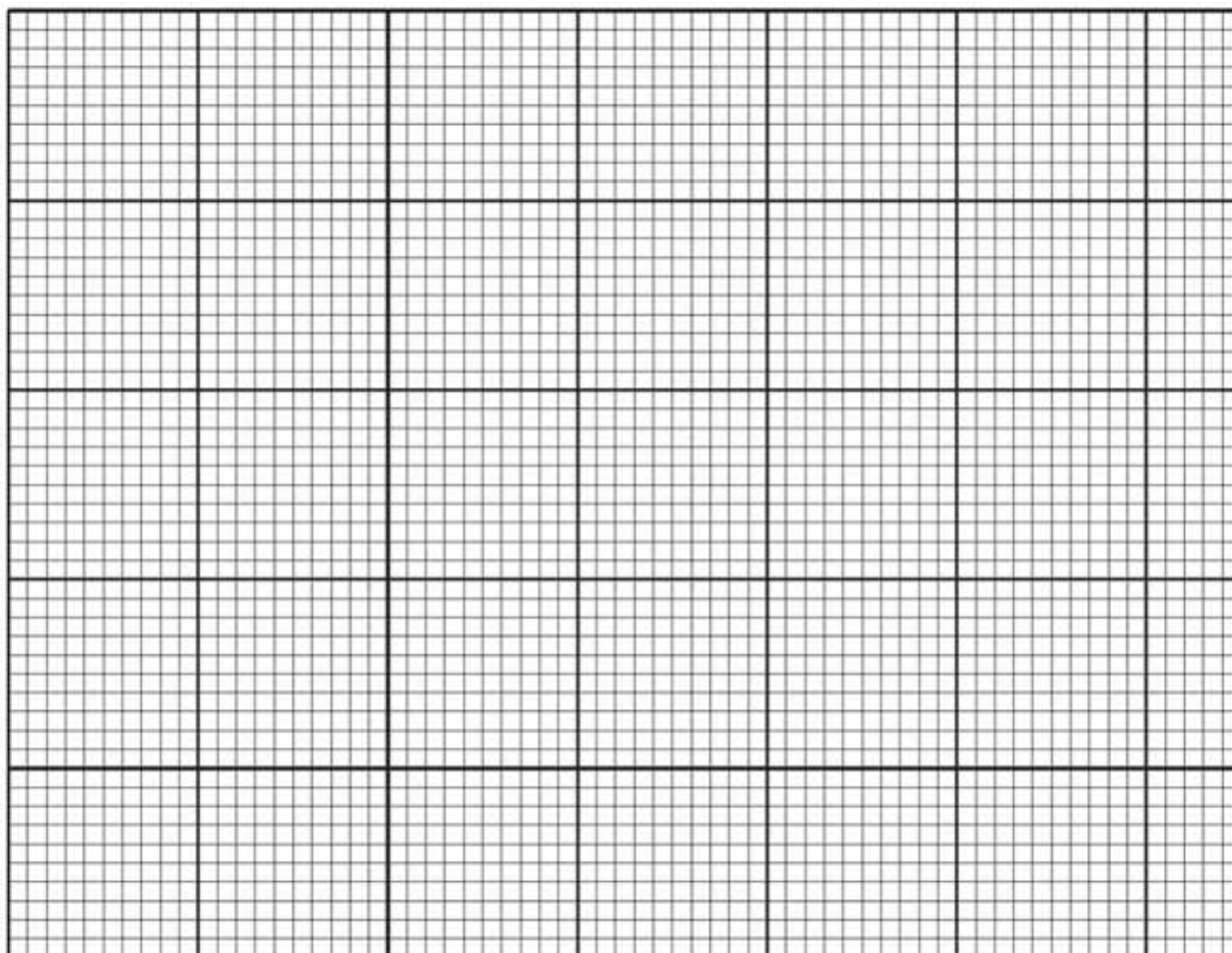
The percentage of starch which reacted with the iodine solution was measured.

The results are shown in Table 1.1.

Table 1.1

volume of iodine solution/ cm^3	percentage of starch which reacted with iodine solution/ %
0.0	0.0
0.5	2.0
1.5	5.0
2.0	36.0
2.5	68.0

- (i)** Plot a graph of the data shown in Table 1.1.
*You will need to consider the answer to **(b)(ii)** before you plot your graph.*



- (ii) Estimate the volume of iodine solution needed for 100% of the starch to be reacted.

Show on your graph how you obtained the volume of iodine solution.

Volume of iodine solution cm³ [1]

- (iii) Explain how the presence of ascorbic acid may affect the use of iodine solution as a test for the presence of starch in different plant tissues.

.....

[2]

- (iv) A plant tissue contains starch and 0.0001 mol dm⁻³ ascorbic acid.

Suggest how you would make sure that the iodine test showed the presence of all the starch (100%).

.....
[1]

- (c) In a separate experiment, starch and amylase were placed into a Visking tubing. The glucose that diffused out of the Visking tubing was tested with Benedict's solution.

- (i) A student wants to study the effect of temperature on the rate of diffusion of glucose through the Visking tubing.

Describe **two** ways to make this experiment valid.

.....

 [2]

- (ii) Explain the effects of increased temperature on the rate of diffusion of glucose through the Visking tubing.

.....

 [2]

[Total: 25]

- 2 Methylene blue stains dead cells blue. Living cells are **not** stained blue so they will appear white or clear.

You are provided with:

- methylene blue solution, **M**, (handle it carefully as it will stain your skin) and
- suspensions of yeast cells, labelled **S1**, **S2** and **S3**.

Each suspension, **S1**, **S2** and **S3** has been heated for ten minutes at 45 °C or 80 °C or 100 °C.

You are required to:

- use the microscope to observe the colour of the yeast cells from **S1**, **S2** and **S3**, after **M** has been added,
- record your observations by using annotated drawings of three yeast cells from each of **S1**, **S2** and **S3** and
- identify the temperature at which each of **S1**, **S2** and **S3** was heated.

1. Label three microscope slides **S1**, **S2** and **S3**.
2. Place **one drop** of **S1** onto slide **S1** and add **one drop** of **M**. Mix carefully using a glass rod. (If **M** comes into contact with your skin rinse with cold water.)
3. Repeat step 2 with **S2** and **S3**.
4. Leave for five minutes.
5. Add a coverslip to each slide.
6. Use the paper towel to dry off any excess liquid around the coverslip.
7. Use the microscope to observe the yeast cells on each slide, then select cells which you can draw and annotate to describe the effect of **M**.

- (a) (i) Prepare the space below and record your observations:

S1



S2

S3

[3]

- (ii) Use your observations to identify the temperature that was used to heat each of the suspensions **S1**, **S2** and **S3**.

Complete the table below.

suspension	temperature/ °C
S1	
S2	
S3	

[1]

- (iii) Besides conducting repeats and replicates, describe how you could modify this investigation to obtain more accurate quantitative measurements.

.....

.....

..... [1]

Fig. 2.1 shows a diagram of a stage micrometer scale that is being used to calibrate an eyepiece graticule.

One division, on either the stage micrometer scale or the eyepiece graticule, is the distance between two adjacent lines.

The length of one division on this stage micrometer is **0.1 mm**.

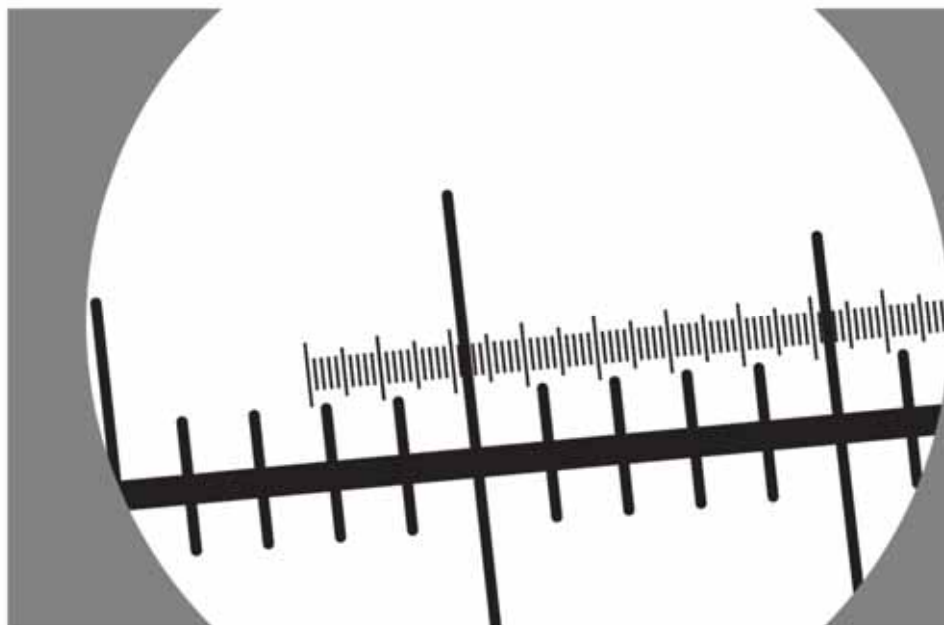


Fig. 2.1

- (b) (i)** Using this stage micrometer, where one division is **0.1 mm**, calculate the actual length of one eyepiece graticule unit using Fig. 2.1 by completing the steps below:

Step 1

1 eyepiece graticule unit = divided by = mm

Step 2

Convert the answer to a measurement with the unit most suitable for use in light microscopy.

.....multiplied by=

[2]

Fig. 2.2 is a photomicrograph of yeast cells.

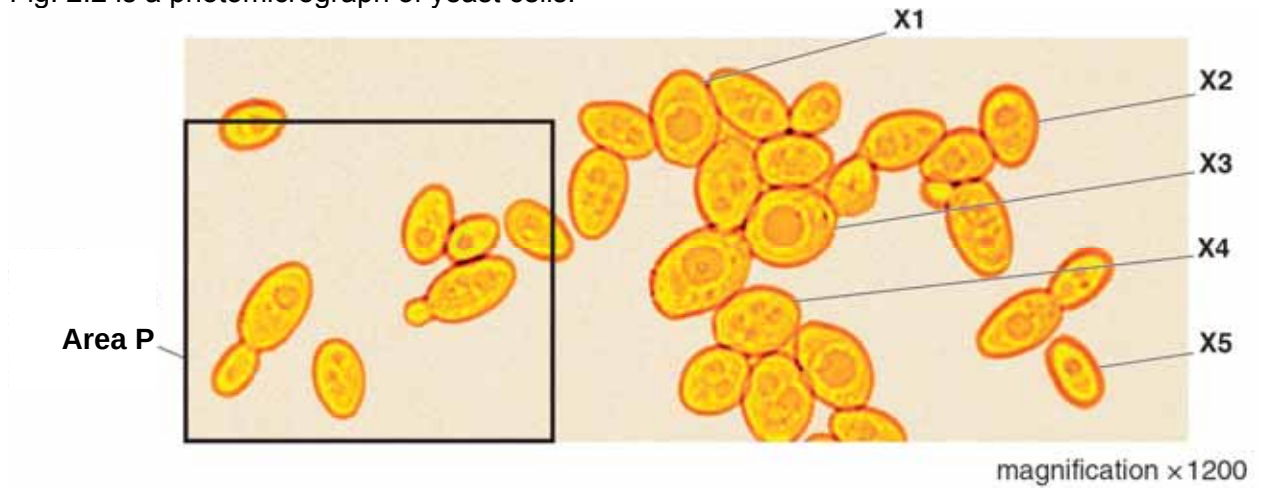


Fig. 2.2

- (c) (i) Use the magnification to calculate the mean maximum actual length, in μm , of yeast cells, **X1**, **X2**, **X3**, **X4** and **X5**.

You may lose marks if you do not show your working or if you do not use the appropriate units.

..... μm

[4]

- (ii) Make a large drawing of the whole cells in area P shown in Fig. 2.2.

[5]

[Total:16]

- 3 Obesity-related diseases such as high blood pressure, heart disease, diabetes and some cancers have been on the rise. High incidence of obesity has been attributed to an unhealthy diet that is high in lipids and sugars.

To combat the issue of a high-fat diet, scientists have been trying to develop drugs that will prevent the absorption of lipids in people's diet. Gastric and pancreatic lipases are enzymes that play a pivotal role in the digestion of dietary lipids.

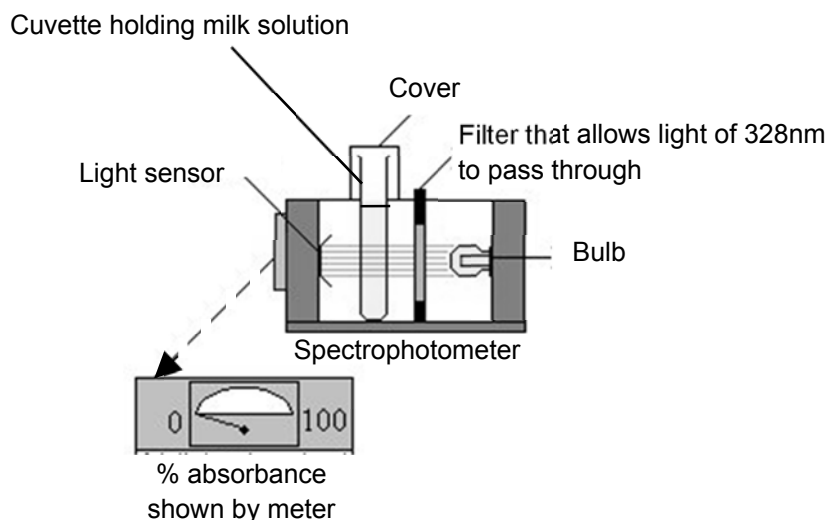
Scientists have recently developed a new drug, Lipobuster, which is a competitive inhibitor of these lipases. In tests performed by the scientists, Lipobuster has been found to have no activity against amylase, trypsin, chymotrypsin and phospholipases.

However, when Lipobuster is administered with lipid-containing foods, it successfully inhibits the hydrolysis of triglycerides. While Lipobuster can help to prevent the absorption of triglycerides, initial studies in mice have shown that high dosages of Lipobuster administered over a long duration resulted in development of tumours.

To ascertain the effectiveness of Lipobuster, scientists have been using full-fat milk in their experiments. Milk is opaque due to the proteins and fats contained within the milk. Thus the effect of Lipobuster on lipase can be investigated by observing its effect on milk opacity.

Opacity of any liquid can be measured using a spectrophotometer, where the reading of absorbance on the spectrophotometer is measured in percent (%). Absorbance is the amount of light that passes through the solution. When all the light passes through the solution, absorbance is zero. In contrast, when no light is able to pass through the solution, absorbance reading would be at 100%.

Note: Lipobuster and lipase are clear colourless solutions.



Using the information given and your own knowledge, design an experiment to determine the minimum concentration of Lipobuster that is needed for lipase to be fully inhibited.

You must use the following:

- full-fat milk
- 5% lipase solution
- 10% Lipobuster solution
- distilled water
- spectrophotometer
- cuvette

You may select from the following apparatus and use appropriate additional apparatus range of common laboratory apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- stopwatch
- thermometer
- syringes
- water trough

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it;
- be illustrated by relevant diagram(s), if necessary, to show for example, the arrangement of apparatus used;
- identify the independent and dependent variables;
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible;
- include layout of results table and graphs with clear headings and labels;
- use the correct technical and scientific terms;
- include reference to safety measures to minimise any risks associate with the proposed experiment.

[Total: 14]

End of Paper